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SHADOW FLEETS, GEOPOLITICAL SHOCKS, BEHAVIOUR & EMISSIONS DIFFERENTIAL

Evidence from the Strait of Hormuz Closure

Abstract

This paper investigates whether geopolitical shocks to maritime chokepoints generate differential behavioural and environmental consequences for shadow and conventional merchant fleets. Exploiting the March 2026 Strait of Hormuz closure as a natural experiment, we apply a difference-in-differences (DiD) framework to vessel-level AIS tracking data to identify the causal effect of an acute geopolitical shock on crossing behaviour across fleet types. We document that the blockade reduced daily Strait crossings by 90.3% for conventional fleet vessels but only 63.3% for shadow fleet operators – a differential driven by the institutional asymmetry between the two groups: conventional vessels face binding war risk insurance constraints that shadow operators, excluded from regulated maritime frameworks, simply do not share. Beyond the crossing differential, we show that shadow fleet vessels carry a structurally higher emissions intensity: a mean annual CO_2 proxy of 26,939 tonnes per vessel against 23,796 for conventional vessels, a 13.2% premium driven by older technology vintages and concentration in tanker and chemical-oil carrier segments. The combination of differential crossing resilience and higher per-vessel emissions intensity means that shadow vessels – representing 5.3% of unique vessels in the sample – account for a disproportionate 5.9% of implied aggregate annual emissions exposure, a share that grows post-closure as shadow vessels constitute an increasing fraction of active crossings. These findings illuminate a structural coherence failure at the intersection of sanctions regimes and maritime climate policy: the institutional mechanisms designed to constrain shadow fleet operations simultaneously insulate their operators from the disruptions those mechanisms generate, with material and widening environmental consequences precisely when geopolitical tensions are highest. We further expand the consideration to potential future shocks and re-routing of regulated fleet, leaving shadow fleet with a resilience and emissions comparative advantage.

Keywords: shadow fleet, environmental economics, maritime emissions, Strait of Hormuz, difference-in-differences, sanctions

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1 Introduction

The global maritime shipping industry sits at an uncomfortable intersection of geopolitics and environmental policy. As sanctions regimes have expanded – particularly following Russia's 2022 invasion of Ukraine and the subsequent acceleration of Western restrictions on both Russian and Iranian oil exports – a so-called "shadow fleet" of older, opaquely owned tankers has grown dramatically. These vessels operate outside the conventional insurance and regulatory frameworks, including the oversight mechanisms of Lloyd's of London and the International Maritime Organization (IMO), that govern mainstream merchant shipping.

This paper investigates a question that has received surprisingly little academic attention: do geopolitical shocks, specifically the threat or reality of strait closures, affect the behaviour and resulting emissions trajectories of shadow and normal merchant fleets differently? The mechanism is asymmetric by design. When war risk premiums spike, conventionally insured vessels face binding financial and regulatory incentives to cease operations in the affected corridor and re-route away from conflict zones. Shadow fleet operators, lacking international insurance coverage and often affiliated with sanctioned states or buyers, face no equivalent constraint. They can continue using shorter, riskier routes and, in this specific case of a closed choke-point like the Strait of Hormuz from which no alternative maritime exit exists, continue to transit at materially higher relative rates. The central question this paper asks is therefore a direct one: do shadow fleet vessels respond less to an acute geopolitical shock than their conventional counterparts, and if so, what are the implied emissions consequences of that differential resilience?

We exploit the abrupt closure of the Strait of Hormuz by Iran in early March 2026 as a natural experiment, applying a difference-in-differences framework to vessel-level AIS tracking data to identify the causal effects of the shock on crossing behaviour and the resulting implied emissions differential across fleet types. The Strait of Hormuz is the world's most strategically important maritime choke-point: approximately 20% of global oil trade passes through its $33km$-wide navigable channel. Unlike disruptions to other corridors where an alternative physical passage exists – such as the Cape of Good Hope re-routing triggered by Houthi attacks on the Bab al-Mandab – a Hormuz closure forecloses maritime exit from the Persian Gulf almost entirely. The behavioural response it generates is therefore unusually clean: a sharp, politically-driven withdrawal by the regulated fleet, against which the shadow fleet's continued operation stands in direct contrast.

The emissions implications of this differential operate through two reinforcing channels. First, the crossing differential itself: as conventional vessels withdraw and shadow vessels remain relatively active, the shadow fleet constitutes a growing share of total remaining transits, and therefore of total crossing-related emissions. Second, a structural emissions intensity gap: shadow fleet vessels are systematically older and concentrated in the tanker and chemical-oil carrier segments that carry the highest per-vessel emissions baselines, meaning that each shadow transit represents a dirtier vessel than its conventional counterpart, independently of the shock. Together, these channels imply that shadow vessels – 5.3% of unique vessels in our sample – account for a disproportionate share of implied aggregate emissions exposure, a share that grows precisely when geopolitical conditions are most severe.

These findings carry implications that extend well beyond the specific case of the Hormuz closure. The core insight is structural: the same institutional features that exclude shadow vessels from the regulated maritime system – the absence of war risk insurance, the opacity of ownership, the indifference to flag state obligations – simultaneously insulate them from the disruptions that system generates. This resilience is likely to manifest not only under geopolitical shocks of the kind studied here, but under a broader class of future disruptions, whether arising from further conflict escalation, evolving sanctions architectures, or climatic and resource-driven shifts that alter the economics of specific trade routes. Addressing it requires not maritime climate regulation or sanctions policy alone, but a dedicated and forward-looking approach to shadow fleet governance.

Our contributions are threefold. First, we offer empirical evidence on the differential response of shadow and conventional fleets to an acute geopolitical shock, using the Hormuz closure as a natural

experiment that allows for causal identification of this differential behaviour. Second, we construct vessel-level emissions proxies grounded in engineering-based fuel consumption models, calibrated to vessel type, size, and age, characterising the structural emissions gap between fleet types independently of the shock itself. Third, we situate these findings within the broader question of shadow fleet resilience, arguing that the institutional features of shadow fleet exclusion are simultaneously the source of their comparative advantage under geopolitical stress – a dynamic with direct implications for how policymakers should think about shadow fleet governance going forward.

The paper is structured as follows. Section 2 surveys the existing literature, covering the origins and growth of the shadow fleet, its environmental footprint, and the strategic significance of the Strait of Hormuz. Section 3 describes our data, classification approach, and emissions proxy construction. Section 4 lays out the conceptual framework and empirical design. Section 5 presents the results of both our DiD and environmental proxy estimations. Section 6 turns to policy implications and recommendations, drawing from our results and the literature reviews. Section 7 discusses limitations and directions for future work. Section 8 concludes.

2 Literature Review & Approach

Throughout this review of the existing literature, we try not only to contextualise our study, but moreover to draw from literature the key specifications, dynamics and risks related to shadow fleets. Most of the papers published on that matter or focused on economic and security matters, with a few deeper analysis on the environmental risks they inherently carry. Even though we'll comment and try to further study the emissions emitted by official VS shadow fleets, the overall characterisation of the latter's formation and functioning is crucial for the other (and what may be the main) contribution of our paper: the resilience and behaviour of ghost vessels under such shocks and constraints. The case of the Hormuz strait is also discussed and highlighted as a critical event for observing these behaviours of a type that may keep occuring in the future. All this motivates our study and brings us to our research question.

2.1 The Origins and Dynamics of the Shadow Fleet

The emergence of shadow fleets predates the current era of sanctions and conflicts, but the geopolitical environment since 2022 has dramatically accelerated its growth. The use of flag-of-convenience and informally operated tankers – changing countries and shell companies affiliations throughout a journey, depending the ports, territorial waters and regulations – as well as the manipulation of their localisation – either by blurring, cutting or spoofing (i.e. faking identity/position) their AIS[1] tracking signals – are key behaviours of these fleets. But one could wonder how and when they are formed. Hereafter, we see that the announce-effect (and more specifically the anticipation) of shocks is a key parameter to understand it, whether it is on the offer side (with standardisations and regulations making some vessels obsolete or too costly to repair) or the demand side (blockades and conflicts such as the Hormuz strait case that foster demand for such cheap, older, less appealing vessels). We later build on these findings to draw policy implications and proposals from our results (Section 6).

Cardoso et al. (2024) document how the G7 oil price cap on Russian exports – designed to limit Russian revenues while maintaining global supply – created powerful structural incentives for the amassing of vessels outside conventional Western insurance and regulatory frameworks. They emphasise the importance of what they call the "surprise effect": the efficacy of new sanctions and regulations depends critically on their anticipation. When market actors foresee impending restrictions, they can pre-position vessels, financing, and ownership structures to circumvent them before enforcement begins, fundamentally undermining the intended economic pressure.

Rodriguez-Diaz et al. (2024) extend this analysis to the global context, demonstrating that the shadow fleet is not just a Russian phenomenon but reflects a broader structural feature of the international oil trade. Their policy recommendations focus on the inadequacy of current enforcement mechanisms: flag state jurisdiction is easily exploited through flags of convenience, and port state control inspections are too infrequent and geographically uneven to provide meaningful deterrence. Crucially, they identify the parallel tracking problem – the simultaneous operation of sanctioned and non-sanctioned oil flows – as a fundamental challenge for any price-cap enforcement regime.

Hilgenstock et al. (2024) provide the most systematic empirical assessment of the Russian shadow fleet's composition and growth trajectory. Their data reveal that the initial build-up was achieved remarkably quickly: within months of the December 2022 price cap implementation, a functional parallel tanker market had been assembled, drawing on older vessels sold by Western ship-owners responding to sanctions risk exposure. Their forward-looking analysis suggests that as the fleet ages further – these are predominantly vessels exceeding 15 years of age – maintenance costs will rise and casualty risks will increase, with implications both for supply security and environmental accident probability.

From a legal and governance perspective, Parlov and Sverdrup (2024) situate the shadow fleet within the broader framework of maritime security law and hybrid warfare. They argue that the

[1] Automatic Identification System

opaque ownership structures, flag-hopping, and AIS manipulation that characterise shadow operations constitute a novel form of maritime security challenge that existing international law frameworks – developed for interstate naval conflict and piracy – are ill-equipped to address.

2.2 Environmental Impact of the Shadow Fleet

The available literature on shadow fleets lacks a deep analysis of their (though often mentioned) environmental impacts. When it does, it focuses primarily on accident risk and pollution exposure (as a statistically characterised event), with fewer considerations of vessel conditions, carbon-intensity, and systematic emissions trajectories.

Knapp et al. (2024) develop a pollution risk exposure assessment framework applied to Portugal and the Baltic Sea, demonstrating that the growth of shadow fleet traffic materially increases expected incident costs at the regional level. Their methodology expresses risk exposure as probabilistic incident cost estimates at the ship level, accounting for vessel age, flag state quality, maintenance standards, and route density. Their results show that a vessel operating without Protection and Indemnity (P&I)[2] club coverage and with deferred maintenance presents an order-of-magnitude higher expected pollution cost per voyage mile than a mainstream tanker.

This literature establishes an important distinction between *flow emissions* – continuous CO_2 and other greenhouse gas emissions from fuel combustion during normal operations – and *stock pollution risk* – tail-risk probability of catastrophic spills from incidents. Shadow fleet vessels are involved in both, but through different mechanisms. The flow emissions channel operates through vessel characteristics themselves: older, less efficient engines, poorer hull maintenance, and lower incentives for slow-steaming. The stock pollution channel operates through casualty risk: ageing vessels with deferred maintenance, operating in conflict-adjacent waters without emergency response coverage, represent materially elevated spill risk. Going even further, a vessel of such a fleet often have fake, unreliable or duped insurers that won't provide any indemnity, leaving countries facing the resulting pollution to pay for the cleaning costs.

Our paper focuses primarily on the flow emissions channel – based on the rich available data we describe in Section 3 – while adopting a novel lens: the differential effect of geopolitical shocks on routing adaptation and therefore total voyage emissions. Indeed, we don't try to replicate estimations of stock pollution risks, but rather to highlight and understand the resilience and behaviour of shadow fleets, along with their environmental implications. Our findings imply that even if shadow vessels are inherently dirtier on a per-mile basis – which the vessel age and maintenance literature strongly suggests – their relative emissions advantage over conventional fleets may grow sharply when geopolitical shocks force the regulated fleet to dramatically extend voyage distances.

2.3 Study Case of the Strait of Hormuz

The Strait of Hormuz has long occupied a central position in energy security analysis. Flanked by Iran to the north and Oman and the U.A.E. to the south, the Strait represents an essential (and for some, irreplaceable direct) choke-point for oil exports from Saudi Arabia, Iraq, Kuwait, Bahrain, Qatar, the U.A.E., and Iran itself. The U.S. Energy Information Administration estimates that approximately 20–21 million barrels per day – roughly 20 percent of global petroleum liquids consumption – transited the Strait in recent years (U.S. EIA, 2024), making any closure or significant disruption an event of global macroeconomic consequence.

Bhattacharya (2026) examines the distributional consequences of Middle Eastern conflict escalation for energy economies, documenting both the direct supply disruption effects and the secondary market distortions that arise when conflict creates incentives for shadow supply chains. His analysis emphasises

[2] A mutual, non-profit maritime insurance that protects shipowners and operators against third-party liabilities and open-ended risks that traditional marine insurers may not cover

that geopolitical conflict rarely eliminates oil flows – it reroutes and obscures them, at economic and environmental cost.

In this case, re-routing isn't obvious and the event is too recent for us to witness significant and long term investments in alternative routes (across the Arabian peninsula through to the Red or Arabian Seas for example) or provision sources (in the short- to medium-term, other countries may increase their production to try to keep up with the global demand and compensate the blocked Arabic supply). Yet, we judge the Hormuz Strait situation an impressive and sudden shock of which type may be increasingly likely to happen and that will surely affect future expectations and behaviours of international oil producers, tanker fleets managers, and policy makers.

To further characterize the studied closure of the Strait of Hormuz (furthered detailed in Section 4), we focus on the start of the American war on Iran and the resulting interdiction of crossing the Strait from Iran, on February 28th-March 1st. An additional American blockade on Iranian exports occurred later in April, but still we choose to focus on this first timestamp as it was a sudden, unannounced and unprecedented shock that constitutes an ideal case study. The framing of the shock is also interesting, as it is mostly formed of a threatening announcement with a few demonstrations and no clearly stated criteria (that later evolved along with declarations and negotiations of both countries). Therefore, we observed (and continue to observe even now with the dual blockade) several ships of various nature and country affiliation crossing the strait. One could argue that this closure is rather a filter from Iran letting only Iranian ship crossing, thus biasing the treatment. Yet we consider the combination of the short considered time-frame, the ambiguous criteria, the diversity of indirectly-related affiliations and the overall war-zone riskiness to generate a wide-enough context for various ships to cross (as we later observe[3]). We could later perform a deeper observation of the evolution of these crossings across a longer period (with several filtering criteria and blockades announcement shocks) with respect to the affiliated countries and there alliances, yet it is too soon to explore such options and we judge it less consistent and robust enough for rigorous statistical observation.

The lasting tensions across the Bab al-Mandab Strait are a good example of an already existing situation that may further escalade in the future. Being the main connection between Asia and Europe (along with the Suez canal), it concentrates in itself 10%-15% of the global trades, and it is another crucial choke-point for hydrocarbons[4]. Since the end of 2023, geopolitical tensions and the threats of Houti rebels attacks has already pushed some shipowners to reroute their ships all the way around the African continent, through the Cape of Good Hope, adding at least 16'000km and millions in cost to each vessel[5], along with additional emissions.

2.4 Motivation & Research Questions

The latter situation offers a great case study to expand on the increased environmental impact for legal vessels and the relative advantage shadow fleet may present in contrast. But before that, we choose to focus on the Hormuz Strait situation, as its net closure (with no possible direct maritime re-routing) and its suddenness offer a great natural experiment to study the expected resilience of ghost vessels through a Difference-in-Differences (DiD) analyses.

We decide to assess the differential response to such a shock between regular and shadow fleets, by aiming at answering the question: *Are shadow fleets relatively less affected by the strait "closure" than legal vessels?* We expect significant results as a strong relative increase of ghost vessels among the remaining fleet that keep crossing the strait. According to literature – but using our own proxy for emissions estimation – we also expect the overall CO_2-emissions per vessel to increase, which we will test in Section 5.2.

[3]Diakun B., Minchin J, Raanan T. (2026), Falakshahi H. (2026), Chia O. (2026)
[4]Dallier et al. (2024), Galluccio C. (2026)
[5]Pieri K. (2026)

3 Data

3.1 Data Sources

Our primary data source is the AXS Marine vessel tracking database, which provides Automatic Identification System (AIS) position reports for commercial vessels operating around the world. In particular, it holds comprehensive vessel coverage on the Strait of Hormuz both pre- and post- the commencement of the 2026 American war on Iran. This includes vessel-level data on ships crossing the strait over a one-month period from February 15th to March 18th 2026 with information on each ship's direction and date of crossing, deadweight tonnage, name and International Maritime Organization (IMO) code, flag, AIS destination, and segment (LPG, tanker, liner, etc.). The database also includes information on ships to the west and east of the Strait of Hormuz by their status as either a sanctioned, ghost, or dark fleet. Any one of these designations is considered to classify a ship as "shadow". Thus, we cross-reference this information with the aforementioned crossing data to create a comprehensive dataset of 1,921 ships that crossed the strait in either direction by their shadow status and other vessel-level characteristics in the period from February 15th to March 18th.

The emissions proxy is constructed using fleet-average fuel consumption estimates from the GreenVoyage2050 emissions calculator, calibrated to the Third IMO Greenhouse Gas Study (IMO, 2014). Vessel build years, used to apply the vintage adjustment, are sourced from the AXS Marine database.

3.2 Shadow Fleet Classification

According to the Atlantic Council, shadow fleets tend to have the following characteristics:

(i) P&I club membership is not with an International Group of P&I Clubs member;

(ii) the vessel has undergone a ship-to-ship transfer in international waters within the past 12 months;

(iii) the vessel has a documented history of AIS signal manipulation (gaps, spoofing, or implausible position jumps).

In general terminology, however, there is no official definition of "shadow vessel". Thus, the terms shadow fleet, dark fleet, ghost fleet, or sanctioned fleet are all often used interchangeably. Typically, they tend to be disproportionately old compared to other merchant vessels, often transport sanctioned cargo, have opaque ownership, lack maritime insurance as per the above-listed, and frequently conceal their movements through spoofing and other such practices. Oftentimes, the different terms are used to distinguish between these various behaviours (Braw, 2026). For the purposes of this study, we use the singular term "shadow fleet" to refer to all such vessels, and the dummy variable for shadow fleet status takes the value of one for ghost, sanctioned, and dark fleets.

3.3 Emissions Estimation

Vessel-level CO_2 emissions are estimated using a two-stage proxy construction based on fleet-average fuel consumption data from the GreenVoyage2050's *Fleet and CO2 Calculator*. In the first stage, each vessel is assigned a baseline annual CO_2 figure derived from the GreenVoyage2050 type-and-size cell corresponding to its vessel segment (tanker, dry bulk, liner, LPG, chemical, MPP, etc.) and its deadweight tonnage band. The GreenVoyage2050 tool reports total annual fuel consumption per ship type and size category, derived from the Third IMO GHG Study's fleet-level analysis of fuel consumption shares. We apply the standard conversion factor of 3.114 tonnes of CO_2 per tonne of heavy fuel oil, consistent with IMO GHG Study methodology, to translate fuel consumption into CO_2 emissions.

In the second stage, the baseline is adjusted upward using a vintage multiplier to reflect the well-documented relationship between vessel age and emissions intensity. Older vessels tend to embody

less efficient engine technology, face weaker Energy Efficiency Design Index (EEDI) requirements – which apply to newbuilds from 2013 onwards – and are more likely to exhibit degraded hull and engine performance from deferred maintenance. We apply a piecewise multiplier by construction-year band:

$$m_i = \begin{cases} 1.00 & \text{if built} \geq 2013 \\ 1.05 & \text{if } 2005 \leq \text{built} \leq 2012 \\ 1.10 & \text{if } 1997 \leq \text{built} \leq 2004 \\ 1.15 & \text{if built} < 1997 \end{cases} \tag{3.1}$$

Table 1: Summary Statistics: Sample Vessels by Fleet Type

	Shadow fleet		**Conventional fleet**		**Full sample**	
	Mean	Median	Mean	Median	Mean	Median
Vessel characteristics						
Deadweight tonnage (DWT)	76,245	46,346	84,295	53,074	83,870	52,797
Build year	2004	2005	2011	2010	2011	2010
Age (years)	21.8	21	15.0	16	15.4	16
Vintage distribution (%)						
Built 2013 or later	4.1	–	39.1	–	37.3	–
Built 2005–2012	34.2	–	41.9	–	41.6	–
Built 1997–2004	58.9	–	15.4	–	17.5	–
Built before 1997	2.7	–	3.7	–	3.6	–
Segment composition (%)						
Tanker	57.5	–	27.1	–	28.8	–
Chemical / chemoil	38.4	–	12.7	–	14.2	–
Dry bulk	0.0	–	21.5	–	20.4	–
Liner	0.0	–	16.4	–	15.6	–
Other	4.1	–	22.2	–	21.0	–
Crossing-level observations						
Unique vessels (N)	73	–	1,309	–	1,382	–
Total crossings	103	–	1,818	–	1,921	–
Share of crossings (%)	5.4	–	94.6	–	100	–

Table 1 reveals three features of the sample that are central to the analysis. First, shadow fleet vessels are substantially older: 58.9 percent were built between 1997 and 2004, compared to 15.4 percent of conventional vessels, and only 4.1 percent of shadow vessels were built after 2013 – the year from which EEDI requirements apply to newbuilds – against 39.1 percent of the conventional fleet. This compositional difference motivates the vintage multiplier applied in the emissions proxy that will be explained in the following section. Second, the shadow fleet is heavily concentrated in tanker and chemical-oil carrier segments (95.9 percent of shadow vessels), whereas the conventional fleet is spread across dry bulk, liner, MPP, and LPG segments as well. Since tankers and chemical carriers carry the highest GreenVoyage2050 baselines, this compositional skew further will elevate the shadow fleet's mean proxy. Third, despite representing only 5.3 percent of unique vessels in the sample, shadow fleet vessels account for 5.4 percent of crossings, indicating a slightly higher crossing frequency per vessel – consistent with their continued operation through the Strait after the March 1st closure.

4 Methodology

4.1 Conceptual Framework

The key insight motivating our design is that shadow and normal fleets face fundamentally different institutional constraints. Normal fleet vessels operate within the regulated maritime system, comply with IMO standards, are subject to international regulations regarding spoofing, and must be underwritten by maritime insurance. This means that when Iran's declared exclusion zone rendered Hormuz transits acutely dangerous, these institutional constraints translated directly into their behaviour in the sea. As geopolitical tensions escalated and wartime risks spiked insurance premiums, operators of compliant vessels faced both financial and regulatory incentives to cease crossings. Shadow fleet vessels, by contrast, operate outside these institutional frameworks. They carry no conventional war risk insurance, sail under flags of convenience with minimal oversight, and face no equivalent regulatory pressure to halt operations in response to the declared threat.

This asymmetry motivates our empirical strategy, which exploits the commencement of the US-Iran war and the subsequent blockade as a quasi-natural experiment to identify the differential effects of an acute geopolitical shock on shadow and normal merchant fleet vessels. We treat this March 1st shock as a treatment that differentially affects the two groups. Normal merchant fleet vessels are impacted through the insurance and regulatory channels, while shadow fleet vessels are largely unaffected by such mechanisms. The difference-in-differences estimator isolates this differential response by comparing the change in crossing rates for normal fleet vessels before and after March 1st against the corresponding change for shadow fleet vessels over the same period. The coefficient on the interaction term $Post_t \times Shadow_s$ captures the extent to which shadow fleet crossing rates diverged from normal fleet rates following the shock, net of any common time trends affecting both groups equally.

The core identifying assumption is that, absent the blockade, both fleet types would have continued crossing the Strait at similar rates. We empirically verify this parallel-trends assumption in the pre-period by plotting daily crossings for shadow and normal fleets over the full sample period, as presented below. The plot shows both normal and shadow fleets to exhibit stable trends in crossings per day for their respective fleet types, with no evidence of divergence prior to the March 1st shock. Following the geopolitical shock of the war and the subsequent blockade on March 1st, the two series diverge sharply, with normal fleet crossings collapsing compared to pre-period levels, while shadow fleet crossings remain relatively stable. This supports the underlying parallel trends assumption, with the divergence being the treatment effect our empirical strategy quantifies.

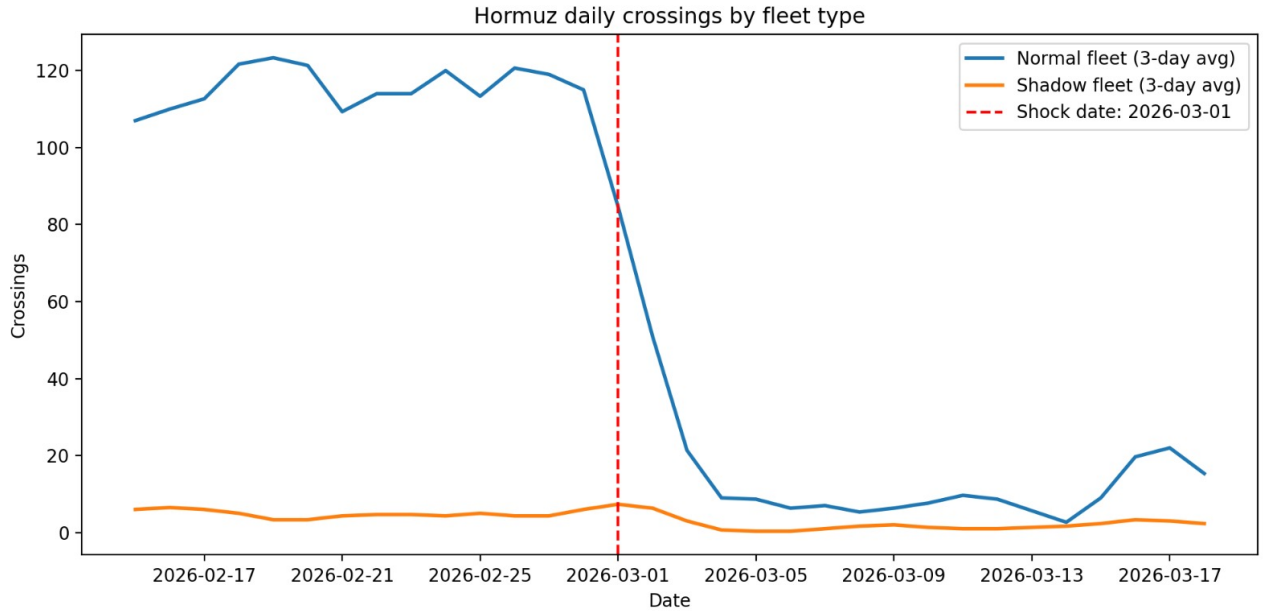


Figure 1: Three-day moving average of daily crossings for normal and shadow fleet types.

Here, it must be noted that the data only contains information on ships that successfully crossed the strait. Thus, our entire sample is affected by the treatment of the geopolitical shock, with the DiD estimator measuring the differential reaction to the treatment between the two subgroups. The ideal experiment would have also included a control group formed by ships that attempted (or intended) to cross but failed to do so. This would have allowed us to measure how being a shadow vessel affects the fulfilment of an intention to cross. However, this suffers from two key limitations. Firstly, our data does not observe all ships that intended to or attempted to cross, but only ships that successfully crossed. Secondly, it is not clear how one can determine whether a vessel aims to cross the Strait of Hormuz or not. While declared AIS destinations can be considered as an indicator in this regard, shadow vessels are defined by their frequent misreporting of such intentions, which would make such a proxy unreliable. Thus, we retain the DiD specification proposed here, which is both feasible and more relevant to the effect of interest we aim to measure – the differential effect of institutional constraints on the reactions of shadow versus conventional fleet vessels to a geopolitical shock.

The environmental dimension of our analysis rests on the premise that shadow and conventional vessels differ not only in their institutional constraints but also in their baseline emissions intensity. As Section 2.2 establishes, the flow-emissions channel operates through vessel characteristics: shadow fleet vessels are systematically older, carry lower-grade fuel, and face weaker incentives for energy efficiency improvements. In our sample, the average shadow vessel is 21.8 years old compared to 15.0 years for non-shadow vessels, and 93.1 percent of shadow vessels in the crossing sample are tankers or chemical-oil carriers – vessel types that carry the highest per-vessel annual CO_2 baselines under the GreenVoyage methodology.

These structural differences mean that, holding routes constant, shadow vessels generate higher annual emissions per vessel than their conventional counterparts. Our emissions proxy captures this baseline asymmetry through a type-size-vintage framework: the baseline CO_2 assignment reflects vessel type and deadweight tonnage, while the vintage multiplier adjusts for technology-vintage differences correlated with EEDI compliance. The proxy is therefore an emissions exposure measure – the annual CO_2 a vessel of given type, size and age would be expected to produce under typical operating conditions – rather than a voyage-specific calculation.

4.2 Empirical Strategy

Our primary empirical specification takes the following difference-in-differences (DiD) form:

$$\text{Crossings}_{st} = \beta_0 + \beta_1 \cdot \text{Post}_t + \beta_2 \cdot \text{Shadow}_s + \beta_3 \cdot (\text{Post}_t \times \text{Shadow}_s) + \varepsilon_{st} \tag{4.1}$$

Crossings_{st} is the daily count of Strait crossings for fleet type $s \in \{\text{shadow, conventional}\}$ on day t. Post_t is an indicator for whether the day is before or after the March 1st, 2026 closure. Shadow_s is an indicator for the shadow fleet, taking a value of 1 for the shadow fleet observations for each day and 0 otherwise. And finally, β_3 is the coefficient of interest, capturing the differential change in crossings for shadow vessels relative to the conventional fleet following the closure.

Given that the outcome variable is a non-negative integer count, we re-estimate the primary specification using a Poisson regression:

$$\mathbb{E}[\text{Crossings}_{st}] = \exp(\gamma_0 + \gamma_1 \cdot \text{Post}_t + \gamma_2 \cdot \text{Shadow}_s + \gamma_3 \cdot (\text{Post}_t \times \text{Shadow}_s)) \tag{4.2}$$

Here, analogous to the OLS specification provided in equation (4.1), γ_3 is the main variable of interest, but expressed as a multiplier rather than a level difference. The Poisson specification relies on an additional assumption that the variance must equal the mean. As our data exhibits overdispersion (Table 2), this assumption does not fully hold. Thus, we also estimate a negative binomial specification. However, given that this yields almost identical and similarly significant results, and OLS does not assume any relationship between the mean and the variance, we adopt the DiD estimator in equation (4.1) as our preferred specification. Across all three specifications, the effect of interest

remains positive and statistically significant, confirming that our result is not sensitive to the choice of estimator.

Table 2: Dispersion Summary Statistics by Group

Group	**Mean**	**Variance**	**Var-Mean Ratio**
Normal fleet – pre	115.50	127.04	1.10
Normal fleet – Post	11.17	128.50	11.51
Shadow fleet – Pre	5.00	4.00	0.80
Shadow fleet – Post	1.83	3.09	1.68

To quantify the structural emissions differential between fleet types, we construct a vessel-level emissions proxy $\widehat{E}_i$. For each vessel i in the sample, the proxy combines a GreenVoyage 2050 baseline annual CO_2 estimate – assigned on the basis of vessel segment s and deadweight tonnage band d – with the vintage multiplier m_i that adjusts for technology-era differences:

$$\widehat{E}_i = \text{Baseline } \text{CO}_2^{s,\,d} \times m_i \tag{4.3}$$

This proxy captures cross-vessel heterogeneity in emissions intensity arising from vessel type, carrying capacity, and technology vintage. It does not capture variation in actual utilisation rates or voyage-specific speed choices; these limitations are discussed in Section 7.2. To assess the validity of the emissions proxy, we regress the vessel-level proxy on the three variables used in its construction: log-transformed deadweight tonnage (DWT), vessel segment fixed effects, and vessel age (years since build). The regression yields an R^2 of 0.827 ($F = 384$, $n = 1{,}382$ unique vessels), indicating that these three variables jointly explain 82.7% of the cross-sectional variation in the proxy. Full results are reported in Appendix B.

The proxy is then used to construct a fleet-type average for shadow and conventional vessels respectively:

$$\bar{E}_g = \frac{1}{N_g} \sum_{i \in g} \widehat{E}_i, \quad g \in \{\text{shadow, conventional}\} \tag{4.4}$$

where N_g is the number of unique vessels in group g. The implied emissions differential is then defined as:

$$\Delta\bar{E} = \bar{E}_{\text{shadow}} - \bar{E}_{\text{conventional}} \tag{4.5}$$

A positive $\Delta\bar{E}$ indicates that the average shadow fleet vessel is associated with higher annual CO_2 emissions than its conventional counterpart, consistent with the structural age and type-composition differences documented in Table 1. The emissions differential is not a causal estimate of the effect of shadow fleet status on emissions per se – it reflects the observable characteristics of vessels that sort into each fleet type – but it provides a quantitative basis for assessing the environmental stakes of the differential crossing behaviour identified in Section 5.2.

5 Results

5.1 Hormuz Differential Analysis Regression Results

Table 3: Regression Results

Variable	DiD	Poisson	Negative Binomial
Intercept	115.500***	4.749***	4.749***
	(2.166)	(0.025)	(0.126)
IRR (Intercept)	—	115.500	115.500
$\mathbf{Post_t}$	−104.333***	−2.336***	−2.336***
	(2.888)	(0.075)	(0.181)
IRR ($Post_t$)	—	0.097	0.097
$\mathbf{Shadow_s}$	−110.500***	−3.140***	−3.140***
	(3.063)	(0.122)	(0.213)
IRR ($Shadow_s$)	—	0.043	0.043
$\mathbf{Post_t \times Shadow_s}$	101.167***	1.333***	1.333***
	(4.084)	(0.224)	(0.323)
IRR ($Post_t \times Shadow_s$)	—	3.793	3.793
α (dispersion)	—	—	0.213***
	—	—	(0.064)
N	64	64	64
R^2	0.971	—	—
Pseudo R^2	—	0.886	0.253

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. OLS coefficients in levels (ships/day). Poisson and Negative Binomial coefficients in log points. Incidence Rate Ratio (IRR) = exp(coefficient).

The table above presents the key regression results for all three regressions, which are significant at the 1% level across all specifications. The full regression results for each of the three specifications are provided in the Appendix. The coefficient on $Post_t$ confirms a sharp and statistically significant decline in normal fleet crossings following the March 1st blockade, with the DiD specification estimating a reduction of 104.3 ships per day. This is equivalent to a 90.3% decline from the pre-period baseline of 115.5 daily crossings. The key coefficient of interest is $Post_t \times Shadow_s$, which captures the differential response of shadow fleet vessels relative to the normal fleet following the blockade. Across all specifications, this coefficient is positive and significant at the 1% level, indicating that shadow fleet crossings declined substantially less than normal fleet crossings after March 1st. The DiD estimate of 101.167 implies that shadow fleet crossings fell by 3.166 ships per day on average after the blockade. This represents a 63.32% decline compared to the 90.33% decline in crossings experienced by normal fleets. The figure presented below showcases the DiD fitted values illustrating this treatment effect. Furthermore, the DiD specification achieved an R^2 of 0.971, indicating that the model almost entirely explains the variation in daily crossings. This reflects the dominance of the March 1st shock and the shadow versus normal fleet distinction in driving crossing behavior over the sample period.

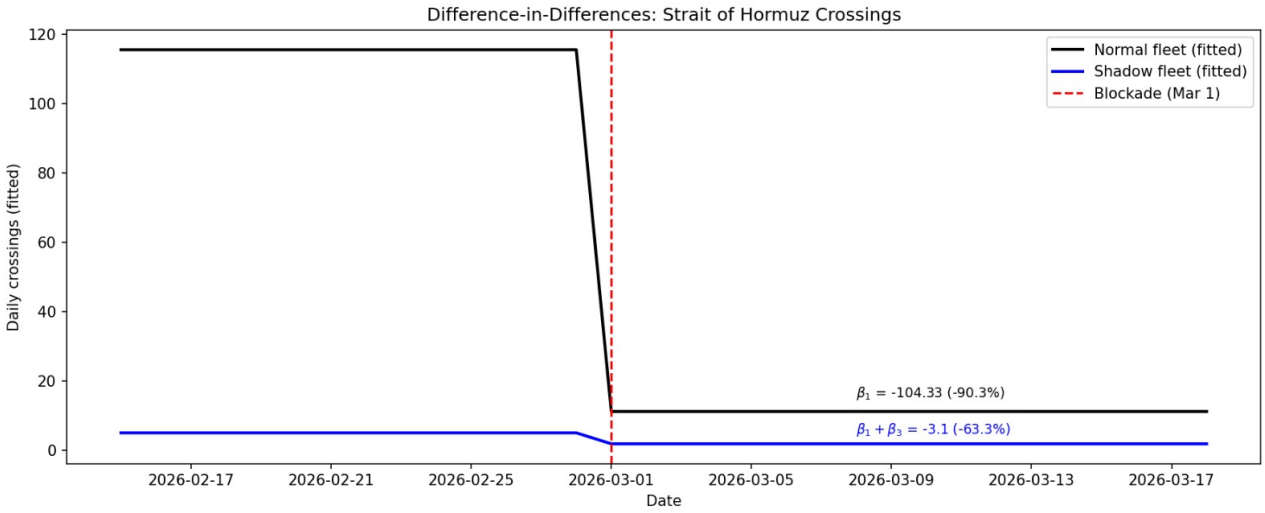


Figure 2: Difference-in-Differences fitted values: Strait of Hormuz Crossings

Both the Poisson and negative binomial specifications yield identical coefficients, with the main effect of interest displaying a corresponding IRR of 3.793. This indicates that shadow fleet post-period crossings were approximately 3.8 times higher than would be expected had shadow vessels responded to the shock in a similar manner as the normal fleet. The subsequently implied percentage changes give essentially the same results as the DiD specification, further validating our empirical strategy. While the significance of the dispersion parameter confirms the presence of overdispersion in the data, motivating the use of a negative binomial regression, this nevertheless yielded identical estimates to the Poisson specification. Although standard errors increased in the negative binomial specification as compared to the Poisson, as is expected due to it accounting for additional variance, the coefficients nonetheless remain significant at the 1% level. The consistency of the effect of the interaction term across all three specifications indicates that our main finding is robust to the choice of estimator. The findings support the conclusion that shadow fleet vessels were substantially insulated from the disruption caused by the blockade, consistent with our expectations given their exclusion from the conventional insurance and regulatory frameworks that govern normal fleet operations.

5.2 Implied Emissions Differential

Table 4 reports the implied annual CO_2 emissions differential between shadow and conventional fleet vessels, computed using the vessel-level proxy described in Section 3.3.

Table 4: Implied Annual CO_2 Emissions Differential by Fleet Type

	Shadow fleet	**Conventional fleet**	**Differential**
Mean annual CO_2 proxy (tonnes)	26,939	23,796	+3,143
Median annual CO_2 proxy (tonnes)	23,057	16,915	+6,142
Standard deviation	17,856	16,641	–
Mean proxy ratio (shadow / conv.)		1.132	
Median proxy ratio (shadow / conv.)		1.363	
N (unique vessels)	73	1,309	–

Notes: Annual CO_2 proxy constructed as described in Section 3.3. Differential is shadow minus conventional. Median ratio uses group medians.

The results confirm the theoretical prediction: shadow fleet vessels carry a structurally higher emissions profile than their conventional counterparts. The mean annual CO_2 proxy is 26,939 tonnes per vessel for shadow operators, compared to 23,796 tonnes for conventional vessels – a mean differential

of 3,143 tonnes, representing a 13.2% premium. The median differential is larger still: 23,057 tonnes versus 16,915 tonnes, a gap of 6,142 tonnes (36.3%). The wider median gap reflects the right-skewed distribution of the conventional fleet, which includes many smaller, newer vessels pulling its mean upward while its median remains lower.

This structural emissions premium is driven by two reinforcing factors documented in Table 1. First, the vintage composition of the shadow fleet is substantially older: 58.9% of shadow vessels were built between 1997 and 2004 – the vintage band attracting a 10% upward multiplier – compared to 15.4% of conventional vessels. Only 4.1% of shadow vessels were built after 2013, against 39.1% of conventional vessels, meaning that the large majority of shadow fleet vessels predate EEDI requirements entirely. Second, 95.9% of shadow vessels are tankers or chemical-oil carriers, the segment classes with the highest GreenVoyage baselines, whereas the conventional fleet is diversified across dry bulk, liner, LPG, and MPP segments with lower per-vessel annual baselines.

Taken together, these findings establish that the shadow fleet's higher crossing rate documented in Section 5.1 is compounded by a higher per-vessel emissions intensity. Even holding crossing counts constant, each shadow fleet transit represents a structurally dirtier vessel. The combination of differential crossing resilience and higher baseline emissions intensity implies that the shadow fleet's aggregate contribution to total crossing-related emissions is disproportionate to its vessel count: while shadow vessels represent 5.3% of unique vessels in the sample, their higher proxy values mean their share of implied aggregate annual emissions exposure is approximately 5.9% – and this share grows post-closure as shadow vessels constitute a larger fraction of remaining active crossings.

6 Policy Implications

6.1 Coordination of Climate and Sanctions Policies

Our findings highlight the need to further examine the interaction between geopolitical sanctions and conflicts with climate and environmental policies. Indeed, the mechanisms that regulate conventional fleet behaviour – insurance exclusions, flag state obligations, ETS compliance – create route-choice incentives that the shadow fleet does not face. When geopolitical shocks activate these mechanisms at scale, the resulting differential in voyage distances translates into a material emissions asymmetry.

Of course, these behavioural responses are here motivated most specifically by military threats any no one should incite vessels to take such risks. But more broadly, this has implications for how we evaluate the environmental performance of maritime climate policy instruments. The International Maritime Organisation (IMO)'s Carbon Intensity Indicator (CII) and the EU Emissions Trading System (ETS) extension to shipping are both designed to impose costs on excessive fuel consumption and emissions. But these instruments apply only to vessels within their regulatory reach. If the shadow fleet grows – as it has done consistently since 2022 – an increasing share of total maritime oil transport emissions falls outside the policy perimeter.

The interference between such environmental regulations and geopolitical sanctions and conflicts should be taken into consideration and anticipated for future estimations and policy design. The coherence problem takes two forms. First, there is a *coverage gap*: shadow fleet vessels are not subject to EU ETS obligations, do not carry the insurance that would expose them to Lloyd's war risk surcharges, and operate under flags that do not enforce IMO regulations. Second, there is an *interaction effect* of the kind we document here: when geopolitical shocks trigger rerouting responses from the regulated fleet, the differential in total voyage emissions between shadow and conventional operations can reverse, even if the shadow vessel is dirtier per mile. Even though this is not the case here, it could be for a scenario such as the one detailed in Section 6.4.

6.2 Shadow Fleet Resilience

Our results speak directly to the resilience question that motivates much of the shadow fleet literature. A 63% decline in shadow fleet crossings following a major strait closure indicates that even shadow operators respond to extreme geopolitical risk. However, the substantially smaller response relative to the 90.3% conventional fleet decline confirms that the absence of insurance coverage, far from deterring shadow operation, breaks down the concern to a raw risk vs benefit analysis for the shadow fleet operators facing significantly lower costs (older, cheaper uninsured vessels escaping several taxes and credit schemes).

From a sanctions effectiveness perspective, this resilience has two implications. First, geopolitical escalation does not terminate sanctioned oil flows – it merely attenuates them, at the cost of increased maritime safety risk. Second, conflict-zone operations become a relatively larger share of the remaining oil trade precisely when conflict risk is highest, raising the expected probability of a major tanker incident in conflict-adjacent waters.

6.3 Addressing Shadow Fleet Specifications

In light of the insights from the literature review (Section 2), we list hereafter potential approaches to address key features of the shadow fleet.

6.3.1 Fleet formation channelling

We saw that even before conflicts like the one studied here, the anticipation of both regulations and sanctions fostered a new parallel market for older vessels. Hence, channelling and choking the

inflow of tankers to the shadow fleet could be crucial challenge. Even though one can't prevent other countries to build their own shadow fleet, a coalition like the EU or the IMO members could stop their ageing fleet from feeding the shadow one. Options like accompanying and incentivising the renovation, maintenance or even the dismantlement of ageing tankers could be a supporting approach. Forbidding the selling of vessels unfitted for travelling under current regulations (and thus enforcing dismantlement or reconditioning) could be a more constraining one.

The question of how can we lose track of sold vessels should also be investigated, as it could shed light of parallel markets and other alternatives for identifying ships independent of their ownership.

6.3.2 Tracking improvement

This part is obviously easier said than done, but is a crucial and central gear to implementing most policies described here and in the literature. Whether it is to hold accountable shipowners or even just target and immobilise ghost vessels, regulators and actors need to be able to recognise such agents. The easier identification & tracking will get, the less latitude shadow fleets will have and the less economically attractive such illegal activities will become. The ideal turning point being the reverse break-even one, where investing in ghost vessels will no longer be profitable.

This kind of information stake has two components: procurement and sharing. The former relies in technological challenges and tools, to better track vessel signals like the AIS one, and potentially coupling it with others like satellite visualisation. The latter is more organisational, as any identification of illegal fleets should be shared with all actors that could get in contact with it – from ports, suppliers and buyers to law enforcement institutions of partnering countries.

6.3.3 Private actors involvement

To enforce identification or banishment of vessels, a regulator should consider delegating some responsibilities and duties to other partners and private actors. For example, it could ask from commercial partners for insurances of no-trading nor hosting of shadow vessels when negotiating. It can more easily impose such regulations to its own ports and actors, even when they trade abroad.

But more broadly, it would be smart to investigate how legitimate shipowners, port managers and insurers (like the P&I Club) could be incentivised to engage in regulating activities, such as reporting shadow vessels, ostracising them or making it increasingly difficult to operate as an unidentified or unregistered actor. Like in many sectors, insurers already became control agents of their clients. There are therefore natural interest for private actors to ensure their market is reliable and monitored that a regulator could exploit.

6.3.4 Environmental responsibility enforcement

We mentioned the limited responsibility shadow vessels take in cases of accidents and stock spillings, leaving concerned countries to deal with the pollution costs. Holding the shipowner accountable in this case and enforcing reparation or payment for it would be a difficult but impacting mechanism. Not only would it limit casualties of such polluting spillings while protecting the concerned population from (part of) the economic cost related to it, but it would moreover further increase the risk and potential costs shadow fleet operators face, making their activity even less attractive. The absence of insurance becoming a too great economical risk again is probably a very strong lever to incite actors compliance.

Though this relies again on the challenge of identifying and tracking down a shipowner through multiples shell companies or negotiation with flagged-countries, it comes back to our considerations for better tracking but also private actors involvement. Indeed, countries that allow shadow vessels to navigate under their flag could face more intense pressure from coalitions like the EU or the IMO members. Information sharing between operators, ports and countries could also allow for better and faster identification of ownership.

6.4 Projections: Red Sea and Future Conflict Scenarios

The Hormuz closure is not an isolated event but rather a sharp demonstration of a general mechanism: any maritime choke-point closure creates the differential re-routing incentive we document. The Houthi attacks on commercial shipping in the Red Sea beginning in late 2023 provide an earlier iteration of the same dynamic, and so do the historical tensions around the Suez Canal. With the growing tensions between the U.S. and China, and the increasing tendency for protectionism and interventionism, we could fear such shocks (whether they are blockades, tariff barriers, conflicts or even just threats) even across the Panama canal, the Strait of Malacca or the Turkish Straits. Our framework predicts that future episodes of choke-point disruption will systematically generate similar differential emissions consequences, with the magnitude proportional to the rerouting distance and the share of trade already carried by shadow vessels – which may increase with every iteration of such disruptions as they grow appealing.

Globally, this highlights the need for overall anticipation of the evolutive commercial, economical and even political behaviours under rising risks and constraints related to geopolitical, climatic, resource- or demand-driven shortages and shocks. In other terms, the more critical the global context will become, the more comparatively advantageous some illegal and impactful behaviours can become, accelerating the spinning of a vicious cycle. For example, one could draw a comparison between the rising need for air-conditioning or fossil fuel energies to face more intense short term temperature shocks, with the relative advantage of shadow fleets described here that may represent an ever higher share of the commercial vessels. We saw here that having a relative advantage under strong constraints can completely shift the power balance, leading to quasi-monopolies for illegal actors. That is why we stress the crucial need to anticipate it and understand the characteristics and behaviours of those fleets, both when facing future shocks and designing ambitious policies.

6.5 Policy Recommendations

The earlier discussed dichotomy around sanctions and their efficacy isn't the only concern with regard to shadow fleet. Current and future potential geopolitical (or even environmental) shocks and conflicts may favour such behaviour as well. Therefore, our recommendations and focus here are not just on adapting the sanctions and environmental regulations, but more globally on the functioning of the overall international maritime trade, for a better control that could ensure better identification and responsibilisation of shadow fleets, indirectly strengthening the regulations, by erasing the comparative disadvantage for legal ships and making illegal maritime practice less economically attractive.

Drawing on our analysis and the broader literature, we identify three priority areas:

(1) Institutional coordination. International coordination between sanctioning and regulating authorities (on all security, economic and climate topics) and field actors will be crucial for coherent and globally applied policies. This will require information-sharing, joint discussions and design of cross-referenced policies from institutions such as the U.S. Office of Foreign Assets Control (OFAC), the EU Sanctions Directorate, Lloyd's war risk committees, the IMO (ideally including their Marine Environment Protection Committee) and other UN bodies. A joint working group with a mandate to assess the environmental consequences of geopolitical instruments, and reversely the pertinence of environmental regulation constraints on actors affected by geopolitical conflicts, would represent a meaningful institutional innovation. Coordination could also be conducted on the field, with enforcement bodies delegated by the joint committee to patrol in international waters, track suspicious vessels and perform controls in ports of partner countries.

(2) Extending the regulatory perimeter. The regulatory perimeter of maritime climate instruments should be extended to encompass shadow fleet vessels through secondary jurisdiction mechanisms. Port state control authorities in major destination ports have the legal authority to deny access to vessels that cannot demonstrate compliance with minimum safety and environmental standards. Coordinated enforcement of existing port state control powers, combined with diplomatic pressure on flag states with high shadow fleet registrations, could meaningfully reduce the regulatory gap without requiring new multilateral agreements. Addressing the laxity of countries that allow flagging without control could be applied at the demand side, by enforcing control at arrival port. Refusing the delivery and even seizing the ship and stock could make sure a vessel would not recidivate, dissuade other actors and economically pay for the intensified searching costs. Non-derogative ID check at any partnering port (potentially supported by the later mentioned monitoring tools) could be done based on criteria like the ones we used to characterise shadow fleet from the AXS data set (documented history of AIS signal, invalid P&I club membership and insurance validity, flag state on the OFAC sanctions list or changed recently/multiple times, history of a ship-to-ship transfer).

(3) Transparency and monitoring. For any regulation or sanction to be effective, an increase in identification and tracking efforts is necessary. A technological arm would be an investment in satellite-based AIS surveillance to better monitor vessels and detect irregularities and behaviours like spoofing or ship-to-ship transfers. An organisational arm would be the reinforcement of an international transparency registry for vessel beneficial ownership – analogous to the Financial Action Task Force (FATF) recommendations for financial sector beneficial ownership and combatting money laundering – would substantially improve the data environment for all policy communities and enforcing actors.

On overall, actions are about allowing for better control and identification of shadow fleet, restricting their scope of action and accessible ports, and enforcing their responsibility regarding accidents and pollution (for example by making shipowners and flagged countries responsible for compensation of pollutions and accidents). By limiting the economic opportunities and increasing the risk taking for shadow fleets, we ensure these practices become less attractive. To compensate for the comparative disadvantage legal players face, regulators could either adapt environmental taxes to the situation and/or even increase any tax/fine/sanction as soon as one of the key shadow-criteria is encountered (flag changing, AIS switch-off, invalid insurance).

7 Discussion & Limitations

7.1 Scope and Generalisability

Our analysis is focused on a specific geopolitical event in a specific choke-point over a relatively short time horizon. The Hormuz closure of early 2026 provides a clean natural experiment, but its very clarity – a sudden, total closure enforced by a state actor with the military capacity to impose it – may make it an upper bound on the actors responsiveness. More gradual geopolitical risk events may generate differential re-routing responses, with corresponding emissions consequences. Caution is therefore warranted in extrapolating our point estimates to other conflict scenarios.

At the same time, the mechanism we document is general. Any choke-point, any insurance exclusion zone, and any sanctions regime that differentially affects the regulatory constraint set of shadow versus conventional operators will generate qualitatively similar effects. The Hormuz case provides a clean empirical proof of concept, while the direction of the effect depends on the nature of the considered shock and the induced global change in transportation (in volume and type).

7.2 Emissions Proxy Limitations

Our emissions estimates rely on engineering-based fuel consumption models applied to vessel specifications and AIS-observed speeds. This methodology is standard in the maritime emissions literature (IMO, 2020) and validated against measured fuel consumption data in multiple studies, but it introduces approximation error at the voyage level. In particular, we do not observe actual fuel consumption on board, engine load factors, or the use of speed reduction measures that operators may deploy opportunistically.

Additionally, our analysis focuses on direct CO_2 emissions from fuel combustion and does not capture the full environmental footprint of the differential routing. Shadow fleet vessels typically burn lower-quality fuel oil, implying additional SOx, NOx, and black carbon emissions that our framework does not quantify. The nature of the cargo and the higher casualty risk and spill probability associated with conflict-zone operation are also outside our current scope.

Because of the focus on fleet behaviour and the recentness of the event, other indirect and broader environmental implications are neither studied. Indeed one could try to estimate the induced trade-off between maritime reduction through the strait with other routes (including the potential building or improvement of pipelines across the arabian peninsula) and supplies (with costs induced by the sudden increase in production for some countries, and the sudden stop in export for others).

We don't discuss either the impact on global energy consumption and the potential medium- to long-term adaptation in production types. One could argue for example that the increasing risks on petrol supply could accelerate the installation of renewables or nuclear solutions, but we judge this consideration too broad and yet limiting regarding our concern here. Indeed, we want to understand the underlying principles and behaviours of vessels response to such shocks, which may just as well impact other trading goods like resources and technologies that these cleaner energy productions rely on.

7.3 Identification Assumptions

Our difference-in-differences identification rests on the parallel trends assumption: that absent the closure, shadow and conventional fleet crossing rates would have evolved similarly. This assumption is untestable by construction, but we provide visual and statistical evidence of parallel pre-trends in Section 5. Potential threats include differential pre-event information acquisition – if shadow operators were better informed of the imminent closure and began pre-positioning before March 1st – or differential fleet composition changes in the lead-up to the closure. We find no significant evidence of such differential pre-trends, but the short pre-event window limits the power of this test.

The critical geopolitical and evolving context we study – even though we consider it a great natural experiment – bears several limitations we must acknowledge. First, as mentioned in Section 4, the two studied groups (conventional and shadow fleets) are both treated (in the sense of undergoing the same shock), and vessels within each group differs in affiliated country and thus probability of being granted passage (or reversely, directly threatened or attacked). Yet we judge still relevant and the best accessible option using a DiD to assess the response differential between the two groups – as for example, Iran may have both shadow and conventional vessels within its own and allies' fleets – and as an RDD was proved as unsuitable here. Again, we believe the shock to be sudden and ambiguous enough to allow for various responses not purely related to an Iranian filtering. However, we acknowledge hereby the serious limitations borne by the very nature of this shock, that would also concern any other conflict and scenario we mention in Section 6, hence remaining somehow relevant to the scope of our study. Regardless of criteria and affiliations of ships, those shocks do increase the share and attractivity of shadow fleets, along with their induced emissions, which is our main research question here.

7.4 Contribution to the Global Literature

Our paper is merely just a modest empirical contribution to a literature that is still in its early stages, on a rising concern but more specifically on a very recent and still evolving context. The intersection of geopolitical economics, maritime regulation, and environmental policy is under-explored, in part because the data demands of credible identification are high: one needs vessel-level tracking data, fleet classification, and an identifiable policy or geopolitical shock simultaneously. The availability of AIS data and the unfortunate frequency of geopolitical disruptions to major shipping corridors are creating new opportunities for this kind of research, and we hope our approach provides a template for future work.

8 Conclusion

This paper has documented a structural asymmetry at the intersection of geopolitical economics and environmental policy: the mechanisms designed to constrain shadow fleet operations – insurance exclusions, sanctions regimes – simultaneously endow their operators with a comparative resilience to disruptions generated by shocks and conflicts. Exploiting the Strait of Hormuz closure as a natural experiment and applying a difference-in-differences framework, we find that the March 2026 blockade reduced conventional fleet crossings by 90.3% while shadow fleet crossings fell by only 63.3% – a differential driven not by strategic adaptation from shadow operators, but by the institutional constraints that bind their conventional counterparts.

The environmental implications compound this crossing differential. Shadow fleet vessels are structurally older and disproportionately concentrated in tanker and chemical-oil carrier segments, producing a mean annual CO_2 proxy 13.2% higher than the conventional fleet and a median differential of 36.3%. The combination of higher crossing resilience and higher per-vessel emissions intensity means that shadow vessels – 5.3% of unique vessels in our sample – account for approximately 5.9% of implied aggregate annual emissions exposure under normal conditions, a share that grows mechanically post-closure as the conventional fleet withdraws and shadow transits constitute a rising fraction of remaining activity. The emissions gap between fleet types thus widens under geopolitical tension peaks.

These findings carry two levels of implication. At the operational level, they establish the resilience of shadow fleet operators as a structural property of their institutional position rather than a matter of tactics or simple risk tolerance (even though the latter is higher for such operators that face less costs and consequences). The 63% decline in shadow crossings confirms that these operators are not indifferent to geopolitical risk, but their response is attenuated relative to the conventional fleet by approximately 27 percentage points, a gap that reflects the absence of the insurance and regulatory architecture that governs mainstream shipping. At the policy level, the findings reveal a coherence failure in the current architecture of international maritime governance. Sanctions designed as geopolitical instruments, climate regulations designed for the regulated fleet, and insurance markets designed around conventional risk assessment are often developed in isolation from one another. Meanwhile, their interaction produces environmental consequences that no instrument was designed to anticipate or manage.

We do not argue that these findings counsel against sanctions or geopolitical engagement – the strategy behind restricting Russian or Iranian oil revenues is mostly independent of environmental concerns. What we do argue is that policymakers should be aware of those side-effects, that a growing shadow fleet makes them increasingly material, and that the tools adequate to address them are neither purely climatic nor purely geopolitical in character. Extending the regulatory perimeter through coordinated port state control, accelerating beneficial ownership transparency, investing in satellite-based AIS monitoring, and building institutional links between sanctions authorities and maritime climate regulators are the priority areas we identify – not as alternatives to existing policy, but as complements that can reduce the environmental cost of the current architecture without compromising its core rationale.

The Hormuz closure of March 2026 is unlikely to be the last event of its kind. As geopolitical fragmentation deepens and the risks of choke-point disruptions multiply – whether around the Arabian Peninsula, the Straits of Malacca or Turkey, or even the Panama Canal – the dynamics documented here will recur, each iteration potentially reinforcing the shadow fleet's operational foothold and the comparative disadvantage of the regulated maritime system. Environmental economists and maritime governance specialists alike should treat this not as an anomaly to be explained away, but as a structural feature of the current international maritime order – one that demands anticipation, coherent policy design, and a willingness to address shadow fleet resilience on its own terms.

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A Emissions Proxy Validation

A.1 Regression Design

To validate the construction of the emissions proxy, we estimate the following cross-sectional OLS regression on the sample of $n = 1{,}382$ unique vessels:

$$\widehat{\mathrm{CO}_2}_i = \alpha + \beta_1 \ln(DWT_i) + \beta_2 \, Age_i + \sum_k \gamma_k \, \mathbf{1}[Segment_i = k] + \varepsilon_i \tag{A.1}$$

where $\widehat{\mathrm{CO}_2}_i$ is the annual proxy emissions value (tonnes) for vessel i, $\ln(DWT_i)$ is the natural logarithm of deadweight tonnage, Age_i is the number of years elapsed since the vessel's build year as of 2026, and the segment indicators $\mathbf{1}[\cdot]$ capture vessel type fixed effects (tanker, dry bulk, liner, chemical tanker, LPG, and others), with one category omitted as reference.

A.2 Results

Table 5 reports the nested model specifications. Vessel size and segment type together explain 82.3% of cross-sectional variation in the proxy ($R^2 = 0.823$). Adding vessel age increases explanatory power to $R^2 = 0.827$. The marginal gain of age is modest in absolute terms (+0.5 percentage points) because age is partially collinear with the size-type structure already captured by the other regressors. Nevertheless, the age coefficient is precisely estimated at $\hat{\beta}_2 = 157$ tonnes CO_2 per additional year of age (95% CI: [107, 206]; $p < 0.001$), confirming that the age penalty is statistically justified. The full model F-statistic of 384 (15 regressors, $n = 1{,}382$) indicates strong joint significance.

Table 5: OLS Regression: Emissions Proxy on Construction Variables

Specification	**Regressors**	**R^2**	**Adj. R^2**	**F-stat**
(1) Segment only	Segment FE	0.449	–	–
(2) DWT only	ln(DWT)	0.586	–	–
(3) Age only	Age	0.032	–	–
(4) Segment + DWT	Segment FE, ln(DWT)	0.823	0.824	405.1
(5) Full model	Segment FE, ln(DWT), Age	0.827	0.825	384.3
Marginal gain: Age | Segment, DWT		+0.005		
Marginal gain: DWT | Segment		+0.373		

Notes: Cross-sectional OLS on $n = 1{,}382$ unique vessels. Dependent variable is the vessel-level annual CO_2 proxy (tonnes). Heteroskedasticity-robust standard errors not shown; results are qualitatively unchanged. Age coefficient in the full model: $\hat{\beta}_2 = 157$ t CO_2/year (95% CI: [107, 206], $p < 0.001$). Reference segment: *chemoil*.

A.3 Figures

Figure 3 shows the R^2 progression across specifications, and Figure 4 shows mean model residuals by vessel age band after controlling for segment and DWT. Older vessels display systematically positive residuals, confirming that the upward age adjustment is directionally correct.

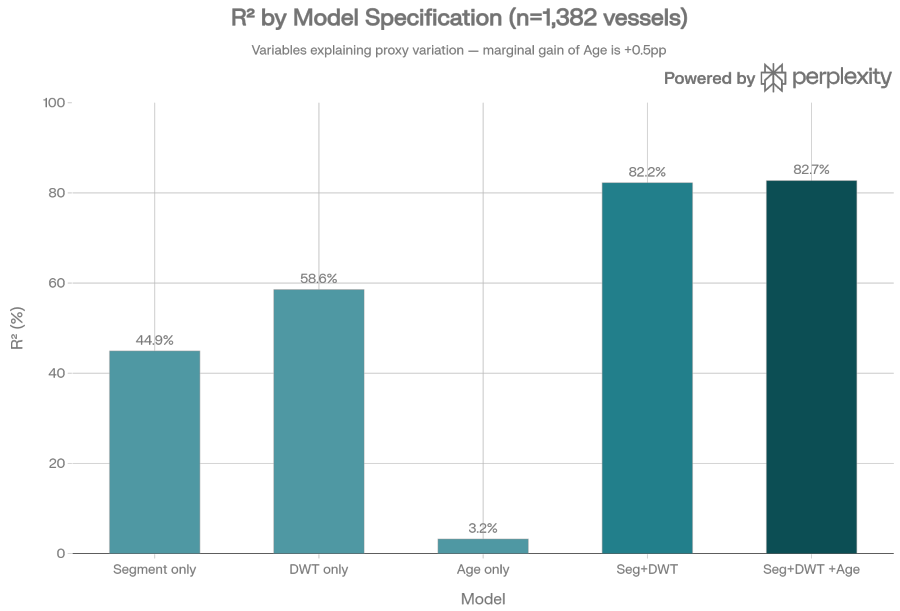


Figure 3: R^2 by model specification (n = 1,382 vessels). Each bar shows the share of cross-sectional variation explained when construction variables are added sequentially.

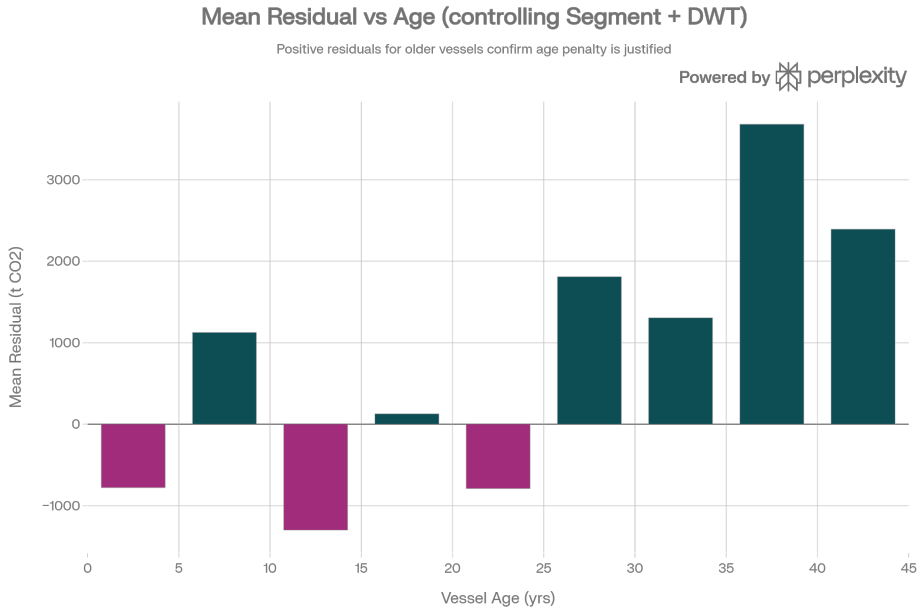


Figure 4: Mean residual by vessel age band after controlling for segment type and DWT. Positive residuals for older vessels confirm the age penalty is directionally justified.

B Full Regression Tables for DiD, Poisson, and Negative Binomial

B.1 DiD Estimation Table

Table 6: DiD Estimation Results

Variable	Coeff.	95% CI Lower	95% CI Upper
Intercept	115.500*** (2.166)	111.168	119.832
Post$_t$	−104.333*** (2.888)	−110.110	−98.557
Shadow$_s$	−110.500*** (3.063)	−116.627	−104.373
Post$_t$ × Shadow$_s$	101.167*** (4.084)	92.997	109.336
$R^2 = 0.971$, Adj. $R^2 = 0.970$, $F(3, 60) = 668.74$, $N = 64$			

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Coefficients in levels (ships/day).

B.2 Poisson Regression Table

Table 7: Poisson Regression Results

Variable	Coeff.	95% CI Lower	95% CI Upper	IRR
Intercept	4.749*** (0.025)	4.701	4.798	115.500
Post$_t$	−2.336*** (0.075)	−2.483	−2.190	0.097
Shadow$_s$	−3.140*** (0.122)	−3.379	−2.901	0.043
Post$_t$ × Shadow$_s$	1.333*** (0.224)	0.894	1.772	3.793
Pseudo $R^2 = 0.886$, Log-Likelihood = −226.15, $N = 64$				

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Poisson coefficients in log points. IRR = exp(coefficient).

B.3 Negative Binomial Regression Table

Table 8: Negative Binomial Regression Results

Variable	**Coeff.**	**95% CI Lower**	**95% CI Upper**	**IRR**
Intercept	4.749*** (0.126)	4.502	4.996	115.500
$\mathbf{Post_t}$	−2.336*** (0.181)	−2.691	−1.982	0.097
$\mathbf{Shadow_s}$	−3.140*** (0.213)	−3.557	−2.722	0.043
$\mathbf{Post_t \times Shadow_s}$	1.333*** (0.323)	0.700	1.966	3.793
α (dispersion)	0.213*** (0.064)	0.089	0.338	1.238
Pseudo $R^2 = 0.253$, Log-Likelihood $= -198.69$, $N = 64$				

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Negative Binomial coefficients in log points. IRR = exp(coefficient).

C Data replication

For replication, understanding and peer-review, we made all our coding files, excel sheets and extracted dataset available here in our GitHub Folder.

It contains, among others files:

- ***main.py*** – main python file to be run for the first (behavioural) analysis of our study, calling all other python files, including
  - ***data_processor.py*** – python file reading and processing the raw AXS data, generating excel files used as input in regressions.
  - ***regression.py*** – python file running the DiD, Poisson and Negative Binomial regressions, saving and exporting the results in combined excel files, generating and saving plots displayed in the report here.
- ***Environmental Analysis.xlsx*** – excel sheet used to compute the emissions proxy, based on included data from AXS and the GreenVoyage2050 Fleet and CO2 Calculator.
- ***README.md*** – text file detailing further the nature and use of the files